

MARIANNA, FL, USA

WMO: 747760

Lat: 30.836N

Lon: 85.184W

Elev: 34

StdP: 100.91

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) -0.4	(d) -10.9	(e) 1.5	(f) 3.2	(g) -8.0	(h) 1.9	(i) 6.4	(j) 9.9	(k) 13.4	(l) 8.9	(m) 13.6	(n) 1.5	(o) 320	(p) 0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	35.8	24.4	34.7	24.4	33.7	24.3	26.7	31.8	26.3	31.2	25.8	30.5	3.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	20.9	28.1	25.1	20.3	27.9	24.7	19.7	27.6	83.6	31.8	81.5	31.2	79.7	30.6	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.3	6.1	-5.4	37.3	1.9	1.7	-6.8	38.5	-8.0	39.5	-9.1	40.5	-10.5	41.7
			-6.7	27.6	2.1	0.6	-8.2	28.0	-9.4	28.4	-10.6	28.7	-12.1	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	10.8	12.8	16.3	19.9	24.3	27.4	28.0	27.9	26.0	20.9	15.2	12.6		
	DBStd	7.28	5.58	4.89	4.50	3.50	2.93	1.90	1.58	1.66	2.39	4.05	4.58	5.07		
	HDD10.0	127	58	24	7	0	0	0	0	0	0	0	9	29		
	HDD18.3	845	239	164	92	24	2	0	0	0	0	22	115	187		
	CDD10.0	3858	84	102	202	298	442	522	557	556	480	339	164	111		
	CDD18.3	1534	7	8	29	72	186	272	299	297	231	103	21	11		
	CDH23.3	14546	24	59	253	657	1768	2707	2988	2932	2078	884	149	46		
	CDH26.7	6154	0	7	43	174	755	1277	1368	1322	907	281	19	1		
(14) Wind	WSAvg	2.4	2.8	2.8	2.8	2.7	2.5	2.2	2.0	2.0	2.2	2.2	2.3	2.5		
(15) Precipitation	PrecAvg	1352	104	116	135	118	83	135	144	152	100	62	84	114		
	PrecMax	1772	206	327	223	290	219	226	300	368	263	174	183	284		
	PrecMin	924	27	20	26	32	28	31	73	56	22	12	20	10		
	PrecStd	279	41	75	66	65	50	57	57	91	57	46	46	71		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	27.8	29.7	31.9	35.5	37.7	36.5	36.9	35.9	32.9	28.6	26.5		
		MCWB	19.5	19.3	18.7	21.5	22.1	23.9	25.4	24.7	23.1	22.7	21.1	20.3		
	2%	DB	23.7	25.2	27.9	30.2	33.4	35.7	35.2	35.2	34.2	31.2	27.0	24.7		
		MCWB	18.3	18.1	18.8	20.4	21.9	24.1	24.9	25.2	23.5	22.2	20.2	20.0		
	5%	DB	21.9	22.8	26.3	28.7	32.2	34.3	34.1	33.9	32.8	29.8	25.0	22.8		
		MCWB	17.9	17.4	18.1	19.4	21.6	24.0	25.0	25.2	23.6	21.2	19.1	19.4		
	10%	DB	19.9	21.1	24.4	27.3	30.9	32.8	32.8	32.8	31.6	28.0	23.1	21.1		
		MCWB	16.7	16.6	17.1	18.9	21.3	23.9	24.9	25.1	23.4	20.7	18.5	18.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.4	22.0	23.5	24.7	26.8	27.3	27.4	26.3	25.5	24.2	22.9		
		MCDB	23.1	24.0	24.8	27.0	29.5	31.9	32.2	33.0	30.7	29.0	25.7	24.3		
	2%	WB	19.9	20.3	21.0	22.6	24.0	26.1	26.7	26.8	25.7	24.6	22.5	21.3		
		MCDB	22.5	23.1	24.1	26.4	28.6	31.3	31.8	31.9	30.1	28.0	24.4	22.9		
	5%	WB	18.7	19.1	20.0	21.7	23.5	25.6	26.2	26.3	25.2	23.7	21.2	20.1		
		MCDB	20.8	21.9	23.6	25.8	28.1	30.8	31.3	31.1	29.7	27.0	23.3	22.0		
	10%	WB	17.3	17.8	18.9	20.7	22.9	25.1	25.7	25.8	24.6	22.7	19.5	18.9		
		MCDB	19.5	20.1	22.9	24.6	27.5	30.1	30.5	30.3	28.8	26.2	22.6	20.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	12.3	13.1	12.9	12.2	10.5	9.9	9.7	10.6	12.4	13.1	11.8		
		MCDBR	12.5	13.4	14.1	13.9	13.2	12.7	11.4	11.1	12.1	13.1	13.3	11.7		
		MCWBR	7.8	7.6	6.6	5.4	4.0	3.5	3.4	3.3	3.8	5.2	7.1	7.4		
	5% WB	MCDBR	10.3	11.0	11.3	10.9	10.6	10.3	10.0	9.8	9.6	9.8	10.2	9.9		
		MCWBR	7.6	7.8	6.7	5.6	3.9	3.6	3.6	3.4	3.5	4.7	6.6	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.339	0.360	0.387	0.440	0.481	0.507	0.500	0.436	0.379	0.351	0.337		
	taud		2.518	2.482	2.432	2.359	2.239	2.187	2.129	2.146	2.305	2.442	2.459	2.507		
	Ebn at Noon		904	923	924	905	855	816	793	794	836	868	866	873		
	Edh at Noon		87	99	111	124	141	148	156	152	125	102	92	85		
(44) All-Sky Solar Radiation	RadAvg		3.08	3.67	4.89	5.94	6.47	6.02	5.83	5.42	4.98	4.40	3.46	2.70		
	RadStd		0.20	0.45	0.38	0.48	0.44	0.47	0.38	0.39	0.36	0.44	0.28	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+163	+108

Nomenclature: See separate page